

DBMS Class Exercise

1. Table Design- Using foreign key and Normalization

- E-R Diagram and reducing to tables
- Table descriptions
- Final Table descriptions with data

Table name: Department

Description: Used to store Department Information

Attribute	Data Type	Constraints
DeptNo	Char(5)	Primary Key/ Not Null
DeptName	Varchar2(35)	
Location	Varchar2(15)	

Table name: Employee

Description: Used to store Employee Information

Attribute	Data Type	Constraints
EmpNo	Char(10)	Primary Key/ Not Null, should start with capital 'E'.
EmpName	Varchar2(30)	
DeptNo	Char(5)	Foreign Key references DeptNo of Department table
Mobile	Char(12)	
Hire_Date	Date	
Job	Varchar2(15)	
EdLevel	SmallInt	
Gender	Char(1)	take only values 'M' or 'F'
DOB	Date	
Salary	Number(10,2)	greater than 0
Bonus	Number(10,2)	
Comm	Number(10,2)	

Table name: Project

Description: Used to store details of projects in the organisation

Attribute	Data Type	Constraints
ProjNo	Char(10)	Primary Key/ Not Null
ProjName	Varchar2(30)	
DeptNo	Char(5)	Foreign Key references DeptNo of Department table
ResEmp	Char(10)	Foreign Key references EmpNo of Employee table
StDate	Date	
EndDate	Date	

Table name: Emp_Proj

Description: Used to store details of employee activity for a project

Attribute	Data Type	Constraints	
ProjNo	Char(10)	Foreign Key references ProjNo of Project table	Primary Key
EmpNo	Char(10)	Foreign Key references EmpNo of Employee table	
EmpTime	Char(10)		

2. Practice SQL Data Definition Language(DDL) commands

3.1 Table creation and alteration

- Create the tables based on the above description
- Add a column called 'Address' of data type 'Varchar' and size= '50' to Employee table
- Drop the above created column.
- Change the size of PhoneNo column in Employee to 10.
- Change the name of the column ResEmp in Project table to RespEmp.
- Destroy the table Emp_Proj and then recreate it .
- Include integrity constraints(if not specified while creation of tables) such as primary key, referential integrity and null constraints given in the previous table descriptions.
- Add the following check constraints
 - Hire date should be greater than date of birth in employee.
 - Project start date should be less than end date .

3. Practice SQL Data Manipulation Language (DML) commands

4.1 Row insertion, deletion and updating

- Insert the data for the tables with data given in the excel

file'Dataset.xls'.

- Create table TESTEMP with the following columns:
EMPNO
LASTNAME
WORKDEPT
HIREDATE
SALARY
BONUS

The data types and null characteristics for these columns should be the same as the corresponding columns in the EMPLOYEE table described at the beginning of the lab exercise.

- Mr. Smith, Ms Baker, and Ms Thomas joined the company. Their data is as follows:

Employee numbers: E0111, E0222, E0333
EmployeeName: SMITH, BAKER, THOMAS
Department number: C01, A00, D11
Date hired: June 6, 2015
Salary: 25,000, 28,000, 33,000
Bonus: 0, NULL, 0

Add the new employees to the TESTEMP table.

- Insert data into the TESTEMP table by copying the appropriate columns in the EMPLOYEE table for those employees that have employee salary less than or equal to 25000.
- Mr. Smith receives a bonus of 500. Make the appropriate data change.
- All employees in department C01 have done a good job. Therefore, they receive a salary increase of 1,000.
- After a short time Mr. Smith leaves the company. Make the appropriate changes.
- Insert data into the TESTEMP table by copying the columns in the EMPLOYEE table where the employees have an employee salary greater than 25000.
- Make the appropriate changes for the following situation: Ammu Anna Pius is assigned to department E01.
- Mrs. Brown joined the company.
For Mrs. Brown enter the following data:
Employee number: E0360
Employee name: BROWN
Department number: D01
Date hired: Date when data is entered
Salary: 45,000
Bonus: unknown
- Delete all rows from the TESTEMP table and drop the TESTEMP

